

20. Rectifiability and singular integrals

Basic singular integrals

The most fundamental singular integral operator is the **Hilbert transform** H on $\mathbf{R}$:

$$Hf(x) = \int \frac{f(t)}{t-x} dt.$$

This formula gives only a formal definition; the integral does not usually exist in the ordinary sense. But for example if f is Lipschitz and integrable over $\mathbf{R}$, it is easy to check that one can define $Hf(x)$ as the principal value for all $x \in \mathbf{R}$,

$$(20.1) \quad Hf(x) = \lim_{\varepsilon \downarrow 0} \int_{\{t: |x-t| > \varepsilon\}} \frac{f(t)}{t-x} dt.$$

This is due to cancellation since one can write for $0 < \varepsilon < \delta$,

$$\int_{\{t: |x-t| > \varepsilon\}} \frac{f(t)}{t-x} dt - \int_{\{t: |x-t| > \delta\}} \frac{f(t)}{t-x} dt = \int_{\{t: \varepsilon < |x-t| \leq \delta\}} \frac{f(t) - f(x)}{t-x} dt$$

and estimate from this that the limit exists. Much more can be said about H . For example, it can be extended as a continuous operator $L^p(\mathbf{R}) \rightarrow L^p(\mathbf{R})$ for $1 < p < \infty$ and if $f \in L^1(\mathbf{R})$ the limit (20.1) exists for $\mathcal{L}^1$ almost all $x \in \mathbf{R}^1$.

In higher dimensions a natural generalization is the so-called Calderón–Zygmund theory. The kernel $1/x$ can be replaced by kernels k in $\mathbf{R}^n$ which have a similar cancellation property and for which $|k(x)|$ behaves like $|x|^{-n}$ near the origin. Then the operator

$$f \mapsto \int k(y-x) f(y) d\mathcal{L}^n y$$

has properties like H . For this theory, see e.g. Stein and Weiss [1] and Stein [1].

Another way to generalize H to the complex plane $\mathbf{C}$ is to keep the same kernel $1/z$, $z \in \mathbf{C}$, but replace the Lebesgue measure by other measures. This leads to the Cauchy transform C_σ of a complex Borel measure σ :

$$(20.2) \quad C_\sigma(z) = \int \frac{d\sigma \zeta}{\zeta - z}.$$

This is important in the theory of analytic functions, as we saw in the previous chapter, because C_σ is analytic in the complement of $\text{spt } \sigma$, and often a given analytic function can be represented as C_σ for some measure σ .

If Γ is a sufficiently nice rectifiable curve, the Cauchy transform C_Γ ,

$$C_\Gamma f(z) = \int_\Gamma \frac{f(\zeta)}{\zeta - z} d\mathcal{H}^1 \zeta,$$

has similar properties on Γ to those of H on $\mathbf{R}$, see e.g. Christ [1] and Murai [2]. For instance, $C_\Gamma: L^2(\Gamma) \rightarrow L^2(\Gamma)$ defines a bounded operator if and only if Γ satisfies the regularity condition $\mathcal{H}^1(\Gamma \cap B(x, r)) \leq Cr$ for $z \in \mathbf{C}$, $r > 0$, by a theorem of David [1], and for any rectifiable curve Γ and $f \in L^1(\Gamma)$ the principal values

$$\lim_{\varepsilon \downarrow 0} \int_{\Gamma \setminus B(z, \varepsilon)} \frac{f(\zeta)}{\zeta - z} d\mathcal{H}^1 \zeta$$

exist for $\mathcal{H}^1$ almost all $z \in \Gamma$. For similar results on singular integrals over surfaces in $\mathbf{R}^n$ with respect to more general kernels, see David [2]–[4], Semmes [1], [3], David and Semmes [1]–[2].

Over purely unrectifiable sets the behaviour of such singular integrals seems to be just the opposite. In this chapter we shall prove some results in this direction. For any positive number s we shall consider the natural s -dimensional generalization of the kernel $1/z$, $z \in \mathbf{C}$, in $\mathbf{R}^n$ defined by

$$(20.3) \quad K_s: \mathbf{R}^n \setminus \{0\} \rightarrow \mathbf{R}^n, \quad K_s(x) = |x|^{-s-1}x.$$

We shall show that if μ is a non-zero non-negative Radon measure on $\mathbf{R}^n$ such that $0 < \Theta_*^s(\mu, x) \leq \Theta^{*s}(\mu, x) < \infty$ and such that the principal values

$$(20.4) \quad K_s \mu(x) = \lim_{\varepsilon \downarrow 0} \int_{\mathbf{R}^n \setminus B(x, \varepsilon)} K_s(y - x) d\mu y$$

exist μ almost everywhere, then s must be an integer and μ must be s -rectifiable. This will also lead to a characterization of rectifiable sets in terms of singular integrals. Here we are using vector-valued integrals which of course can be defined in terms of the coordinate functions.

Symmetric measures

We shall use tangent measures in the same sense as before in connection with densities. It will turn out that the assumption on the existence of principal values will force the tangent measures to be symmetric in the following sense.

20.5. Definition. Let ν be a non-zero Radon measure on $\mathbf{R}^n$. A point $x \in \mathbf{R}^n$ is called a *symmetric point* of ν if

$$\int_{B(x,r)} y \, d\nu y = x\nu(B(x,r)) \quad \text{for all } r > 0.$$

The set of such symmetric points is denoted by $S(\nu)$. If $\text{spt } \nu \subset S(\nu)$, ν is said to be a *symmetric measure*.

Note that this is a very restrictive condition. It says that for any ball B centred at $\text{spt } \nu$ the centre of mass of the restriction $\nu \llcorner B$ is the centre of this ball. Obvious examples of symmetric measures in $\mathbf{R}^2$ are for $0 < c < \infty$, $c\mathcal{L}^2$, $c\mathcal{H}^1 \llcorner L$ for any line L , and

$$\sum_{m \in \mathbf{Z}} (a\mathcal{H}^1 \llcorner (L + 2mv) + b\mathcal{H}^1 \llcorner (L + (2m+1)v))$$

for any line L , any vector $v \neq 0$ orthogonal to L , and any positive numbers a and b . It was shown by Mattila [16] that in $\mathbf{R}^2$ there are no other continuous (i.e. measures ν with $\nu(\{x\}) = 0$ for $x \in \mathbf{R}^2$) symmetric measures. The corresponding result in $\mathbf{R}^n$ is not known, nor is the characterization of discrete symmetric measures in $\mathbf{R}^2$ (in $\mathbf{R}^1$ this is an exercise). In Theorem 20.9 we shall give a partial answer in $\mathbf{R}^n$.

First we give two simple characterizations of symmetry.

20.6. Lemma. Let ν be a non-zero Radon measure on $\mathbf{R}^n$ and s a positive number. Then the following three conditions are equivalent for $x \in \mathbf{R}^n$:

- (1) $x \in S(\nu)$.
- (2) $\int_{B(x,R) \setminus B(x,r)} K_s(y-x) \, d\nu y = 0$ for $0 < r < R < \infty$.
- (3) $\int (y-x) \varphi(|y-x|) \, d\nu y = 0$ for all bounded Borel functions $\varphi: [0, \infty) \rightarrow \mathbf{R}$ such that

$$\int |y-x| \varphi(|y-x|) \, d\nu y < \infty.$$

Proof. Obviously (3) implies (1) and (2). Suppose that (2) holds. Writing

$$\int (y-x) \varphi(|y-x|) d\nu y = \int K_s(y-x) |y-x|^{s+1} \varphi(|y-x|) d\nu y,$$

we note that (3) holds if $t^{s+1}\varphi(t)$ is a characteristic function of an interval in $(0, \infty)$. For a general φ , (3) follows then by a simple approximation. Similarly (1) implies (3). $\square$

Existence of principal values and tangent measures

The following lemma motivates our study of symmetric measures.

20.7. Lemma. *Let $s > 0$. Suppose μ is a finite Radon measure on $\mathbf{R}^n$ such that for μ almost all $x \in \mathbf{R}^n$, $\Theta_*^s(\mu, x) > 0$ and $K_s\mu(x)$ exists (as in (20.4)). Then for μ almost all $a \in \mathbf{R}^n$ every $\nu \in \text{Tan}(\mu, a)$ is symmetric with $0 \in S(\nu)$.*

Proof. Let $\varepsilon > 0$. By the Cauchy criterion for convergence the existence of $K_s\mu(x)$ means that

$$(1) \quad \lim_{0 < \varepsilon < \delta \downarrow 0} \int_{B(x, \delta) \setminus B(x, \varepsilon)} K_s(y-x) d\mu y = 0.$$

Using Egoroff's theorem we find a compact set F such that $\mu(\mathbf{R}^n \setminus F) < \varepsilon$ and the convergence in (1) is uniform for $x \in F$. (We can apply Egoroff's theorem to the sequence

$$\sup_{0 < \varepsilon < \delta < 1/i} \left| \int_{B(x, \delta) \setminus B(x, \varepsilon)} K_s(y-x) d\mu y \right|$$

of μ measurable functions.) Moreover, F can be chosen so that $\Theta_*^s(\mu, x) > 0$ for $x \in F$.

Let $a \in F$ be a μ density point of F , that is,

$$(2) \quad \lim_{r \downarrow 0} \frac{\mu(B(a, r) \setminus F)}{\mu(B(a, r))} = 0.$$

Since by Corollary 2.14 this is true for μ almost all $a \in F$, it suffices to prove the claim at such a point a .

Let $\nu = \lim_{i \rightarrow \infty} c_i T_{a_i, r_i} \# \mu \in \text{Tan}(\mu, a)$ and let $x \in \text{spt } \nu$. We show that $x \in S(\nu)$; the argument to show that $0 \in S(\nu)$ is similar. Since $x \in \text{spt } \nu$, (2) implies as in the proof of Lemma 14.7 (1) that there is a sequence $a_i \in F$ such that $x_i = (a_i - a)/r_i \rightarrow x$. Remark 14.4 (1) yields

$$(3) \quad \limsup_{i \rightarrow \infty} c_i r_i^s \leq \Theta_*^s(\mu, a)^{-1} \limsup_{i \rightarrow \infty} c_i \mu(B(a, r_i)) < \infty.$$

If $0 < r < R < \infty$ and $\nu(\partial B(x, r)) = \nu(\partial B(x, R)) = 0$, we can apply Theorem 1.19, the uniform convergence of (1) on F , and (3) to obtain

$$\begin{aligned} \int_{B(x, R) \setminus B(x, r)} K_s(y - x) d\nu y &= \lim_{i \rightarrow \infty} \int_{B(x_i, R) \setminus B(x_i, r)} K_s(y - x_i) d\nu y \\ &= \lim_{i \rightarrow \infty} c_i \int_{B(x_i, R) \setminus B(x_i, r)} K_s(y - x_i) dT_{a_i, r_i} \# \mu y \\ &= \lim_{i \rightarrow \infty} c_i \int_{B(a_i, r_i R) \setminus B(a_i, r_i r)} K_s(T_{a_i, r_i}(y) - x_i) d\mu y \\ &= \lim_{i \rightarrow \infty} c_i r_i^s \int_{B(a_i, r_i R) \setminus B(a_i, r_i r)} K_s(y - a_i) d\mu y = 0. \end{aligned}$$

Since $\nu(\partial B(x, r))$ can be positive for at most countably many radii r , we deduce the symmetry of x by a simple approximation and Lemma 20.6. $\square$

20.8. Remark. The above proof also gives the following statement: if μ is a symmetric measure such that for some positive numbers c and d , $cr^s \leq \mu(B(x, r)) \leq dr^s$ for all $x \in \text{spt } \mu$ and $r > 0$, then the conclusion of Lemma 20.7 holds at all points $a \in \text{spt } \mu$.

Symmetric measures with density bounds

Next we shall derive information about symmetric measures with bounded density ratios.

20.9. Theorem. Let $s > 0$. Suppose there exists a symmetric measure ν on $\mathbf{R}^n$ such that for some positive numbers c and d

$$(1) \quad cr^s \leq \nu(B(x, r)) \leq dr^s \quad \text{for } x \in \text{spt } \nu \text{ and } r > 0.$$

Then s is an integer and ν is a constant multiple of $\mathcal{H}^s \llcorner V$ for some s -plane V .

Proof. We may assume $0 \in \text{spt } \nu$. Let $t > 0$. The symmetry, (1) and Lemma 20.6 imply

$$\int (y - x) e^{-t|y - x|^2} d\nu y = 0 \quad \text{for } x \in \text{spt } \nu,$$

whence

$$\int (y - x) e^{-t|y|^2} e^{2ty \cdot x} d\nu y = 0.$$

Using a Taylor expansion for $e^{2ty \cdot x}$, we obtain from this

$$\int (y - x) e^{-t|y|^2} (1 + 2ty \cdot x + t^2(y \cdot x)^2 g(ty \cdot x)) d\nu y = 0,$$

where $|g(u)| \leq e^{2|u|}$. Let

$$I(t) = \int e^{-t|y|^2} d\nu y.$$

Dividing by $I(t)$ and using the symmetry of ν at 0, we have

$$(2) \quad -x + 2t \int y e^{-t|y|^2} y \cdot x d\nu y / I(t) \\ + t^2 \int (y - x) e^{-t|y|^2} (y \cdot x)^2 g(ty \cdot x) d\nu y / I(t) = 0.$$

We show that the last term tends to zero as $t \downarrow 0$, in fact its norm is bounded by $C\sqrt{t}$. Since $e^{-t|y|^2} |g(ty \cdot x)| \leq e^{-t|y|^2 + 2t|y \cdot x|} \leq b e^{-t|y|^2/2}$ for $0 < t \leq 1$, with b depending on x , this follows from the following estimates:

$$(3) \quad C^{-1}t^{-s/2} \leq I(t) \leq Ct^{-s/2} \quad \text{for } t > 0,$$

$$(4) \quad \int |y|^3 e^{-t|y|^2/2} d\nu y \leq Ct^{-(3+s)/2} \quad \text{for } t > 0,$$

where C is a positive and finite constant depending only on s , c and d . We prove the right hand inequality in (3); the other two follow by similar arguments. Using (1) we estimate

$$I(t) = \int_{B(t^{-1/2})} e^{-t|y|^2} d\nu + \sum_{i=0}^{\infty} \int_{B(2^{i+1}t^{-1/2}) \setminus B(2^i t^{-1/2})} e^{-t|y|^2} d\nu \\ \leq \nu(B(t^{-1/2})) + \sum_{i=0}^{\infty} e^{-2^{2i}} \nu(B(2^{i+1}t^{-1/2})) \\ \leq dt^{-s/2} \left(1 + \sum_{i=0}^{\infty} e^{-2^{2i}} 2^{s(i+1)} \right) \leq Ct^{-s/2}.$$

Defining linear maps $A_t: \mathbf{R}^n \rightarrow \mathbf{R}^n$ for $t > 0$ by

$$A_t x = 2t \int y e^{-t|y|^2} y \cdot x \, d\nu y / I(t),$$

we consequently have

$$(5) \quad \lim_{t \downarrow 0} A_t x = x \quad \text{for } x \in \text{spt } \nu.$$

Let V be the linear subspace of $\mathbf{R}^n$ spanned by $\text{spt } \nu$ and let k be its dimension. Clearly (5) holds for all $x \in V$. Our first goal is to show that $k = s$. By (1), $s \leq k$. By the definition of A_t we have $A_t x = 0$ whenever x belongs to the orthogonal complement of V . Hence by (5) the linear maps A_t converge to the orthogonal projection P_V as $t \downarrow 0$. Thus we have for their traces $(\text{Tr } A_t = \sum_{i=1}^n (A_t e_i) \cdot e_i)$

$$(6) \quad \text{Tr } A_t = 2t \int e^{-t|y|^2} |y|^2 \, d\nu y / I(t) \longrightarrow \text{Tr } P_V = k,$$

as $t \downarrow 0$.

Suppose now that $s \neq k$. Then $s < k$ and by (6) there are $u > s$ and $t_0 > 0$ such that $\text{Tr } A_t > u$ for $0 < t < t_0$.

Note that

$$\text{Tr } A_t = -2tI'(t)/I(t),$$

whence for $t < t_0$, $-2tI'(t) > uI(t)$, which gives that the derivative of h ,

$$h(t) = t^{u/2} I(t),$$

is negative. Thus h is strictly decreasing on $(0, t_0)$ and so

$$t^{s/2} I(t) = t^{(s-u)/2} h(t) \longrightarrow \infty \quad \text{as } t \downarrow 0.$$

This contradicts (3) and proves that $s = k$. In particular, s is an integer.

We have now shown that $\text{spt } \nu$ is contained in an s -plane, so we may assume ν is a measure on $\mathbf{R}^s$. We must prove that $\nu = c\mathcal{L}^s$.

We first show that $\text{spt } \nu = \mathbf{R}^s$. Otherwise we can find an open ball $U = U(b, r)$ and a point a such that $U \cap \text{spt } \nu = \emptyset$ and $a \in \text{spt } \nu \cap \partial U$. Let $\lambda \in \text{Tan}(\nu, a)$. Then by 14.7 (4) and 20.8 λ is symmetric, it satisfies (1) (in place of ν) and, with $e = (a - b)/|a - b|$,

$$0 \in \text{spt } \lambda \subset \{x \in \mathbf{R}^s : x \cdot e \geq 0\}.$$

The symmetry then implies that $\text{spt } \lambda \subset \{x : x \cdot e = 0\}$, see Exercise 3, which leads to a contradiction with (1). So $\text{spt } \nu = \mathbf{R}^s$.

Let $\varphi: \mathbf{R} \rightarrow [0, \infty)$ be continuously differentiable with compact support. Define

$$f(x) = \int \varphi(|x - y|) d\nu y \quad \text{for } x \in \mathbf{R}^s.$$

Then for $i = 1, \dots, n$, $x \in \text{spt } \nu = \mathbf{R}^s$,

$$\partial_i f(x) = \int (x_i - y_i) |x - y|^{-1} \varphi'(|x - y|) d\nu y = 0$$

by Lemma 20.6. Hence f is constant. Approximating the characteristic functions of the balls $B(r)$ with such functions φ we find that ν is uniformly distributed. By Theorem 3.4, $\nu = c\mathcal{L}^s$ as required. $\square$

Existence of principal values implies rectifiability

We can now conclude that principal values do not usually exist for fractal-type measures:

20.10. Theorem. *Let $s > 0$. Suppose there exists a finite non-zero Radon measure μ on $\mathbf{R}^n$ such that for μ almost all $x \in \mathbf{R}^n$, $0 < \Theta_*^s(\mu, x) \leq \Theta^{*s}(\mu, x) < \infty$ and $K_s \mu(x)$ exists. Then s is an integer and μ is s -rectifiable.*

Proof. By Lemma 20.7 the tangent measures of μ at μ almost all points are symmetric. By Lemma 14.7 they also satisfy condition (1) of Theorem 20.9. Hence, by that theorem, s must be an integer and the tangent measures at μ almost all points are s -flat. Finally, Theorem 16.7 implies that μ is rectifiable. $\square$

20.11. Remarks. The above results are from Mattila and Preiss [1] where somewhat more was proved. Assuming only that $\Theta_*^s(\mu, x) > 0$ and that $K_s \mu(x)$ exists for μ almost all $x \in \mathbf{R}^n$, one can still prove that at μ almost all points all tangent measures are flat, but possibly of different dimensions at different points, see Mattila and Preiss [1]. I do not know if μ must be in some sense rectifiable. In the case $n = 2$, $s = 1$, μ was shown to be 1-rectifiable under these assumptions in Mattila [16]. However, if s is an integer, $0 < \Theta_*^s(\mu, x) < \infty$ and $K_s \mu(x)$ exists for μ almost all $x \in \mathbf{R}^n$, Theorem 5.6 of Preiss [4] together with Mattila and Preiss [1] implies that μ is s -rectifiable.

Under the assumptions of Theorem 20.10 μ is absolutely continuous with respect to $\mathcal{H}^s$. But if we drop the assumption on the finiteness of upper density this is no longer clear even in the case $n = 1$, $s = 1$: it does not seem to be known whether there exists a Radon measure μ on $\mathbf{R}$, singular with respect to $\mathcal{L}^1$, such that $K_1\mu(x)$, the principal value for the Hilbert transform, exists for μ almost all $x \in \mathbf{R}$. Note that $\Theta^1(\mu, x) = \infty$ for μ almost all $x \in \mathbf{R}$ when μ is singular as a consequence of Theorem 2.12.

L^p-boundedness and weak (1, 1) inequalities

We shall now develop some ideas which will yield a converse to Theorem 20.10 and which are interesting in their own right, too. The complete treatment would take us too far into the theory of singular integrals and we will just give references to some parts. First we fix some notation and terminology. In what follows the spaces $L^p(\mu)$ will consist of complex-valued functions.

20.12. Definition. Let m be an integer with $0 < m \leq n$, $0 < C < \infty$ and $k: \mathbf{R}^n \setminus \{0\} \rightarrow \mathbf{C}$ a continuously differentiable function with $k(-x) = -k(x)$,

$$(1) \quad |k(x)| \leq C|x|^{-m} \quad \text{and} \quad |\nabla^j k(x)| \leq C(j)|x|^{-m-j}$$

for $x \in \mathbf{R}^n \setminus \{0\}$, where $\nabla^j k(x)$ stands for the vector whose coordinates are all the j -th order partial derivatives of k . The corresponding truncated kernels k_ε , $\varepsilon > 0$, are defined by

$$k_\varepsilon(x) = \begin{cases} k(x) & \text{for } |x| > \varepsilon, \\ 0 & \text{for } |x| \leq \varepsilon. \end{cases}$$

Let S be a closed subset of $\mathbf{R}^n$ and μ a Radon measure with $\text{spt } \mu = S$ and

$$(2) \quad C^{-1}r^m \leq \mu(B(x, r)) \leq Cr^m \quad \text{for } x \in S \text{ and } 0 < r < d(S).$$

For $\varepsilon > 0$ and $f \in \bigcup_{1 \leq p < \infty} L^p(\mu)$, we define the truncated operators T_ε by

$$T_\varepsilon f(x) = \int k_\varepsilon(x - y) f(y) d\mu y$$

and the corresponding maximal operator T^* by

$$T^* f(x) = \sup_{\varepsilon > 0} |T_\varepsilon f(x)|.$$

Note that $T_\varepsilon f(x)$ is defined since k_ε is bounded and belongs to $L^q(\mu)$ for $1 < q \leq \infty$ because of (1) and (2).

We shall also consider the same operators for complex Radon measures σ on $\mathbf{R}^n$:

$$T_\varepsilon \sigma(x) = \int k_\varepsilon(x-y) d\sigma y,$$

$$T^* \sigma(x) = \sup_{\varepsilon > 0} |T_\varepsilon \sigma(x)|.$$

We shall assume throughout the rest of this chapter that $k, k_\varepsilon, m, C, S, \mu, T_\varepsilon$ and T^* are as above.

Next we state two basic results on the singular integral operators T_ε without proofs.

20.13. Theorem. Suppose that the operators $T_\varepsilon, \varepsilon > 0$, are uniformly bounded on $L^2(\mu)$. This means that there exists $c_2 < \infty$ such that

$$(1) \quad \int |T_\varepsilon f|^2 d\mu \leq c_2 \int |f|^2 d\mu$$

for all $f \in L^2(\mu)$ and $\varepsilon > 0$. Then T^* is bounded on $L^p(\mu)$ for $1 < p < \infty$ and of weak type (1,1), that is, there are constants $C_p < \infty$ such that

$$(2) \quad \int |T^* f|^p d\mu \leq C_p \int |f|^p d\mu \quad \text{for } f \in L^p(\mu), \quad 1 < p < \infty,$$

and

$$(3) \quad \mu(\{x : T^* f(x) > t\}) \leq C_1 t^{-1} \int |f| d\mu \quad \text{for } f \in L^1(\mu) \text{ and } t > 0.$$

20.14. Remarks. (1) If $m = n$ and $\mu = \mathcal{L}^n$ this is one of the basic results of the Calderón–Zygmund theory, see Stein [1] and Stein and Weiss [1]. The same technique can be adapted to this more general situation and even further: Coifman and Weiss [1] develop the theory in homogeneous spaces; any metric space with a Borel measure μ satisfying $\mu(B(x, 2r)) \leq c\mu(B(x, r))$ for all x, r is homogeneous. They only present the corresponding results for the operators T_ε and not T^* , but this follows then by an application of Cotlar's inequality, see Journé [1, p. 56], which we shall also prove in Lemma 20.25. The conditions on k can be

somewhat relaxed, and of course in general metric spaces 20.12 (1) must be replaced by a different condition.

(2) Clearly 20.12 (2) implies that μ is comparable with $\mathcal{H}^m \llcorner S$ and we could work with that measure as well.

(3) The Cauchy kernel $z \mapsto 1/z$, $z \in \mathbf{C}$, and the Riesz kernels $x \mapsto x_i|x|^{-m-1}$, $x \in \mathbf{R}^n$, $i = 1, \dots, n$, are basic examples to which the above results apply.

For $m = n$ and $\mu = \mathcal{L}^n$ the condition 20.13 (1) is most conveniently verified with the help of the Fourier transform, see Stein and Weiss [1] or Stein [1]. But we are now mainly interested in the cases where it holds for $m < n$. The following is a fundamental result.

20.15. Theorem. *Suppose that S is the graph of a Lipschitz function $f: B \rightarrow \mathbf{R}^{n-m}$, where B is a ball in $\mathbf{R}^m$,*

$$S = \{(x, f(x)) : x \in B\},$$

and $\mu = \mathcal{H}^m \llcorner S$. Then 20.13 (1) holds and hence also 20.13 (2) and (3). Moreover, if $f \in L^1(\mu)$, the principal values

$$Tf(x) = \lim_{\varepsilon \downarrow 0} T_\varepsilon f(x)$$

exist for μ almost all $x \in S$.

20.16. Remarks. (1) This result was first proved by Calderón [1] for $n = 2$ and the Cauchy kernel $k(z) = 1/z$ in the case $\text{Lip}(f) < \delta$, where δ is a small absolute constant; recall the discussions from the preceding chapter. The restriction on $\text{Lip}(f)$ was removed by Coifman, McIntosh and Meyer [1]. See also Murai [2] where many later proofs are presented. Perhaps the simplest known proofs are given in Coifman, Jones and Semmes [1]. More general one-dimensional kernels were handled in Coifman, David and Meyer [1]. The m -dimensional Lipschitz graphs and kernels can be reduced to the 1-dimensional case by the method of rotation, see Coifman, David and Meyer [1], David [2, p. 245] and Guzmán [2, § 5.3]. In fact, Theorem 20.15 holds for much more general surfaces S due to results of David and Semmes, see David [4], David and Semmes [1]–[2] and the references given there. For the last statement of Theorem 20.15, see David [4, p. 63].

(2) So after condition (1) of Theorem 20.13 has been established, that general theorem takes care of the rest except the existence of principal

values. In fact, it is not known if the principal values exist μ almost everywhere in the situation of Theorem 20.13 for example for the Riesz kernels. In view of Theorem 20.10 a positive solution would imply that the L^2 -boundedness assumption 20.13 (1) would force S to be m -rectifiable. Combining with Remark 19.18 (5) we could then conclude that Conjecture 19.8 on analytic capacity is true for sets $S \subset \mathbf{C}$ satisfying 20.12 (2) with $m = 1$.

The reason that the almost everywhere existence of the principal values can be verified in Theorem 20.15 is the regularity of S . Using that one can first show by more direct arguments that $Tf(x)$ exists for μ almost all x if f is smooth. Since smooth functions are dense in $L^1(\mu)$, the weak type inequality 20.13 (3) can then be used to extend the almost everywhere convergence to L^1 -functions. For a general S satisfying 20.12 (2) we do not have any functions from which we could start.

A duality method for weak (1, 1)

It is fairly easy to see that once we have 20.13 (3) for L^1 -functions we have it also for complex Radon measures with support in S . But we would like to have this weak type inequality for arbitrary complex Radon measures on $\mathbf{R}^n$ without any restriction on the support. For this we shall use the argument of Verdera [2]. It relies on an elegant duality method which goes back to Uy [1]–[3]. Duality arguments in connection with singular integrals have also been used in Davie and Øksendal [1], Hruščev [1], Murai [2] and Verdera [1]. The presentation below will often be based on that of Christ [1].

We denote by Z^* the complex dual of a Banach space Z , that is, Z^* consists of continuous linear functions $Z \rightarrow \mathbf{C}$. The following form of the Hahn–Banach theorem can be found in Rudin [2, Theorem 3.4].

20.17. Theorem. *Let Z be a Banach space and B_1, B_2 disjoint non-empty convex subsets of Z . Suppose B_2 is open. Then there exists $\lambda \in Z^*$ such that*

$$\operatorname{Re} \lambda(x) > \operatorname{Re} \lambda(y) \quad \text{for } x \in B_1, y \in B_2,$$

where $\operatorname{Re} z$ is the real part of the complex number z .

For a locally compact Hausdorff space X let $C_c(X)$ be the set of continuous functions $\varphi: X \rightarrow \mathbf{C}$ vanishing at infinity, i.e. for every $\varepsilon > 0$ there is a compact set K such that $|\varphi(x)| < \varepsilon$ for $x \in X \setminus K$. Equipped

with the norm $\|\varphi\| = \sup\{|\varphi(x)| : x \in X\}$ it is a Banach space. Let $\mathcal{M}_c(X)$ be the space of complex Radon measures σ on X equipped with the total variation norm $\|\sigma\|$. It is the dual of $C_c(X)$. Let X and Y be locally compact Hausdorff spaces and $L: \mathcal{M}_c(X) \rightarrow C_c(Y)$ a bounded linear operator. The *transpose* of L , $L^t: \mathcal{M}_c(Y) \rightarrow C_c(X)$ is defined by

$$\int L\sigma \, d\tau = \int L^t\tau \, d\sigma \quad \text{for } \sigma \in \mathcal{M}_c(X), \tau \in \mathcal{M}_c(Y),$$

provided such a linear operator L^t exists. (In general the transpose of L has range in the dual of $\mathcal{M}_c(X)$.)

20.18. Theorem. *Let X and Y be locally compact Hausdorff spaces, ν a Radon measure on X and $L: \mathcal{M}_c(X) \rightarrow C_c(Y)$ a bounded linear operator. Suppose that $L^t: \mathcal{M}_c(Y) \rightarrow C_c(X)$ is of weak type (1,1), that is, there is $c < \infty$ such that*

$$(1) \quad \nu(\{x \in X : |L^t\sigma(x)| > t\}) \leq ct^{-1}\|\sigma\|$$

for $t > 0$ and $\sigma \in \mathcal{M}_c(Y)$. Then for any Borel set $B \subset X$ with $0 < \nu(B) < \infty$, there exists a Borel function $h: X \rightarrow [0, 1]$ such that $h(x) = 0$ for $x \in X \setminus B$, $\int h \, d\nu \geq \nu(B)/2$ and $\|Lh\| < 3c$. Here h is identified with the measure $A \mapsto \int_A h \, d\nu$.

Proof. Suppose that this fails for some B . Define

$$B_0 = \{f: X \rightarrow [0, 1] : f \text{ is a Borel function with}$$

$$f = 0 \text{ on } X \setminus B \text{ and } \int f \, d\nu \geq \nu(B)/2\},$$

$$B_1 = \{Lf : f \in B_0\},$$

$$B_2 = \{g \in C_c(Y) : \|g\| < 3c\}.$$

The hypotheses of Theorem 20.17 are satisfied with $Z = C_c(Y)$. Hence there exists $\lambda \in C_c(Y)^* = \mathcal{M}_c(Y)$ such that $\operatorname{Re} \lambda(h) > \operatorname{Re} \lambda(g)$ for $h \in B_1$, $g \in B_2$, that is,

$$\operatorname{Re} \int Lf \, d\lambda > \operatorname{Re} \int g \, d\lambda \quad \text{for } f \in B_0, g \in B_2.$$

The supremum of the right hand side when g runs through B_2 equals $3c\|\lambda\|$, whence

$$(2) \quad \operatorname{Re} \int f L^t \lambda \, d\nu \geq 3c\|\lambda\| \quad \text{for } f \in B_0.$$

We apply the weak type inequality (1) with $t = 2c\|\lambda\|/\nu(B)$ to obtain

$$\nu(\{x : |L^t\lambda(x)| > t\}) \leq ct^{-1}\|\lambda\| = \nu(B)/2$$

so that

$$\nu(\{x \in B : |L^t\lambda(x)| \leq t\}) \geq \nu(B)/2.$$

Letting f be the characteristic function of the Borel set $\{x \in B : |L^t\lambda(x)| \leq t\}$, we have $f \in B_0$ and

$$\left| \int f L^t \lambda \, d\nu \right| \leq t\nu(B) = 2c\|\lambda\|,$$

contradicting (2). $\square$

This can also be reversed: the existence of a function h as above for all compact subsets B of X leads to the weak type inequality (1).

20.19. Theorem. *Let X and Y be locally compact Hausdorff spaces, ν a Radon measure on X and $L: \mathcal{M}_c(X) \rightarrow C_c(Y)$ a bounded linear operator whose transpose L^t maps $\mathcal{M}_c(Y)$ into $C_c(X)$. Then the following two conditions are equivalent:*

(1) *There is $c_1 < \infty$ such that*

$$\nu(\{x \in X : |L^t\sigma(x)| > t\}) \leq c_1 t^{-1} \|\sigma\|$$

for $t > 0$ and $\sigma \in \mathcal{M}_c(Y)$.

(2) *There is $c_2 < \infty$ such that for every compact set $F \subset X$,*

$$\nu(F) \leq c_2 \sup \left\{ \int h \, d\nu : h: X \rightarrow [0, 1] \text{ is a Borel function with } h = 0 \text{ on } X \setminus F \text{ and } \|Lh\| \leq 1 \right\}.$$

Moreover, the least constants c_1 and c_2 satisfy $c_1/8 \leq c_2 \leq 6c_1$.

Proof. If (1) holds Theorem 20.18 yields (2) with $c_2 = 6c_1$. Suppose (2) holds. Let $\sigma \in \mathcal{M}_c(Y)$ and let F be a compact subset of the set $\{x \in X : \operatorname{Re} L^t\sigma(x) > 1\}$. By (2) there exists a Borel function $h: X \rightarrow [0, 1]$ with $h = 0$ on $X \setminus F$, $\|Lh\| \leq 1$ and $\nu(F) \leq 2c_2 \int h \, d\nu$. Then

$$\int h \, d\nu \leq \int h \operatorname{Re} L^t\sigma \, d\nu = \operatorname{Re} \int Lh \, d\sigma \leq \|\sigma\|.$$

Hence for any $\sigma \in \mathcal{M}_c(Y)$,

$$\nu(\{x \in X : \operatorname{Re} L^t\sigma(x) > 1\}) \leq 2c_2 \|\sigma\|.$$

This easily yields (1) with $c_1 = 8c_2$. $\square$

A smoothing of singular integral operators

20.20. Smoothed operators. We shall now return to our setting described in 20.12. We would like to apply Theorem 20.19 to the truncated operators T_ε of 20.12, but there is a problem, that they do not map $\mathcal{M}_c(\mathbf{R}^n)$ to $C_c(\mathbf{R}^n)$. Hence we consider the following smoothed operators. Let φ be a radial C^∞ function on $\mathbf{R}^n$ with $0 \leq \varphi \leq 1$, $\varphi = 0$ on $B(1)$ and $\varphi = 1$ on $\mathbf{R}^n \setminus B(2)$. For $\varepsilon > 0$ define the smooth kernel $\tilde{k}_\varepsilon$ by

$$\tilde{k}_\varepsilon(x) = \varphi(x/\varepsilon) k(x)$$

and the corresponding operator $\tilde{T}_\varepsilon$ by

$$\tilde{T}_\varepsilon f(x) = \int \tilde{k}_\varepsilon(x-y) f(y) d\mu y$$

for $f \in \bigcup_{1 \leq p < \infty} L^p(\mu)$, and

$$\tilde{T}_\varepsilon \sigma(x) = \int \tilde{k}_\varepsilon(x-y) d\sigma y$$

for complex Radon measures σ on $\mathbf{R}^n$. Recall also from 2.18 the Hardy–Littlewood maximal function

$$M_\mu f(x) = \sup_{r>0} \frac{1}{\mu(B(x,r))} \int_{B(x,r)} |f| d\mu$$

for μ measurable functions f , and we also define

$$M_\mu \sigma(x) = M_\mu(|\sigma|)(x) = \sup_{r>0} \frac{|\sigma|(B(x,r))}{\mu(B(x,r))}$$

for complex Radon measures σ .

20.21. Lemma. *If σ is a complex Radon measure on $\mathbf{R}^n$, then for $\varepsilon > 0$, $x \in \mathbf{R}^n$,*

$$|T_\varepsilon \sigma(x) - \tilde{T}_\varepsilon \sigma(x)| \leq 2^{m+1} C^2 M_\mu \sigma(x)$$

where C is the constant of 20.12.

Proof. Since $\tilde{k}_\varepsilon(z) = k_\varepsilon(z)$ for $|z| \leq \varepsilon$ and for $|z| > 2\varepsilon$, and $|\tilde{k}_\varepsilon| \leq |k_\varepsilon|$, we have by 20.12 (1) and (2),

$$\begin{aligned} |T_\varepsilon \sigma(x) - \tilde{T}_\varepsilon \sigma(x)| &\leq \int |k_\varepsilon(x-y) - \tilde{k}_\varepsilon(x-y)| d|\sigma| y \\ &\leq 2 \int_{B(x,2\varepsilon) \setminus B(x,\varepsilon)} |k_\varepsilon(x-y)| d|\sigma| y \\ &\leq 2C\varepsilon^{-m} |\sigma|(B(x,2\varepsilon)) \leq 2^{m+1} C^2 M_\mu \sigma(x). \quad \square \end{aligned}$$

It follows from this lemma and Theorem 2.19 that if T_ε satisfies a weak type (1,1) inequality so does $\tilde{T}_\varepsilon$ and vice versa. We are aiming to apply Theorem 20.19 to transfer a weak type (1,1) inequality from complex Radon measures supported by S to arbitrary complex Radon measures on $\mathbf{R}^n$. One more link is needed for this.

20.22. Lemma. *Suppose that $\tilde{T}_\varepsilon$ is bounded on $L^2(\mu)$. Then for $f \in L^1(\mu) \cap L^\infty(\mu)$,*

$$\|\tilde{T}_\varepsilon f\|_{\mathbf{R}^n} \leq c(\|f\|_{L^\infty(\mu)} + \|\tilde{T}_\varepsilon f\|_S)$$

where $c < \infty$ depends only on m , the constant C of Definition 20.12 and the $L^2(\mu)$ -norm of $\tilde{T}_\varepsilon$.

Here and below

$$\|\varphi\|_X = \sup \{|\varphi(x)| : x \in X\},$$

and $\|f\|_{L^\infty(\mu)}$ is the usual L^∞ -norm of f with respect to μ .

Proof. Let $x \in \mathbf{R}^n \setminus S$ and $\delta = d(x, S)$. Choose $x_0 \in S$ with $|x - x_0| = \delta$. Write $f = g + h$ where $g = f\chi_{B(x_0, 4\delta)}$. Let $y \in B(x_0, 2\delta)$. Suppose $\delta \leq \varepsilon$. Since $\tilde{k}_\varepsilon(z) = k(z)$ for $|z| > 2\varepsilon$ and $|\tilde{k}_\varepsilon| \leq C\varepsilon^{-m}$ by Definitions 20.12 and 20.20 we can estimate

$$\begin{aligned} & |\tilde{T}_\varepsilon h(x) - \tilde{T}_\varepsilon h(y)| \\ & \leq 2C\varepsilon^{-m} \int_{B(x_0, 4\varepsilon)} |h| d\mu + \int_{\mathbf{R}^n \setminus B(x_0, 4\varepsilon)} |k(x - z) - k(y - z)| |f(z)| d\mu z \\ & \leq 2 \cdot 4^m C^2 \|f\|_{L^\infty(\mu)} + \int_{\mathbf{R}^n \setminus B(x_0, 4\varepsilon)} |\nabla k(\xi(z) - z) \cdot (x - y)| |f(z)| d\mu z \end{aligned}$$

where $\xi(z)$ lies on the segment joining x to y . Thus $|\xi(z) - z| \geq |x_0 - z|/2$ for $z \in \mathbf{R}^n \setminus B(x_0, 4\varepsilon)$ and so by 20.12(1) the last summand can be

estimated by

$$\begin{aligned}
 & 2^{m+1}C|x-y|\int_{\mathbf{R}^n\setminus B(x_0,4\varepsilon)}|x_0-z|^{-m-1}|f(z)|\,d\mu z \\
 & \leq 6\cdot 2^mC\delta\sum_{i=2}^{\infty}\int_{B(x_0,2^{i+1}\varepsilon)\setminus B(x_0,2^i\varepsilon)}|x_0-z|^{-m-1}|f(z)|\,d\mu z \\
 & \leq 6\cdot 2^mC\delta\|f\|_{L^\infty(\mu)}\sum_{i=2}^{\infty}(2^i\varepsilon)^{-m-1}\mu(B(x_0,2^{i+1}\varepsilon)) \\
 & \leq 6\cdot 4^mC^2\delta\|f\|_{L^\infty(\mu)}\varepsilon^{-1}\sum_{i=2}^{\infty}2^{-i} \\
 & \leq 6\cdot 4^mC^2\|f\|_{L^\infty(\mu)}.
 \end{aligned}$$

Thus we have, provided $\delta \leq \varepsilon$,

$$|\tilde{T}_\varepsilon h(x) - \tilde{T}_\varepsilon h(y)| \leq c_1\|f\|_{L^\infty(\mu)}.$$

If $\varepsilon < \delta$ we integrate over $B(x_0, 4\delta)$ and $\mathbf{R}^n \setminus B(x_0, 4\delta)$ and get the same estimate using the fact that $h = 0$ on $B(x_0, 4\delta)$.

For g we have by 20.12 (1) and 20.20,

$$\begin{aligned}
 |\tilde{T}_\varepsilon g(x)| & \leq C\int_{B(x_0,4\delta)}|z-x|^{-m}|f(z)|\,d\mu z \\
 & \leq C\delta^{-m}\|f\|_{L^\infty(\mu)}\mu(B(x_0,4\delta)) \leq c_2\|f\|_{L^\infty(\mu)},
 \end{aligned}$$

since $|z-x| \geq \delta$ for $z \in \text{spt } \mu$. Thus

$$\begin{aligned}
 (1) \quad |\tilde{T}_\varepsilon f(x)| & \leq c_3\|f\|_{L^\infty(\mu)} + |\tilde{T}_\varepsilon h(y)| \\
 & \leq c_3\|f\|_{L^\infty(\mu)} + \|\tilde{T}_\varepsilon f\|_S + |\tilde{T}_\varepsilon g(y)|
 \end{aligned}$$

for $y \in S \cap B(x_0, 2\delta)$. By Hölder's inequality and the L^2 -boundedness of $\tilde{T}_\varepsilon$,

$$\begin{aligned}
 \int_{B(x_0,2\delta)}|\tilde{T}_\varepsilon g(y)|\,d\mu y & \leq (\mu B(x_0,2\delta))^{1/2}\|\tilde{T}_\varepsilon g\|_{L^2(\mu)} \\
 & \leq c_4\delta^{m/2}\|g\|_{L^2(\mu)} \leq c_5\delta^m\|f\|_{L^\infty(\mu)} \\
 & \leq c_6\mu(B(x_0,2\delta))\|f\|_{L^\infty(\mu)}.
 \end{aligned}$$

Averaging the inequality (1) with respect to y and μ over $B(x_0, 2\delta)$ we have from this

$$|\tilde{T}_\varepsilon f(x)| \leq c(\|f\|_{L^\infty(\mu)} + \|\tilde{T}_\varepsilon f\|_S)$$

as required. $\square$

20.23. Lemma. Suppose that $c < \infty$,

$$(1) \quad \int |\tilde{T}_\varepsilon f|^2 d\mu \leq c \int |f|^2 d\mu \quad \text{for } f \in L^2(\mu)$$

and

$$(2) \quad \mu(\{x : |\tilde{T}_\varepsilon f(x)| > t\}) \leq ct^{-1} \int |f| d\mu \quad \text{for } t > 0, f \in L^1(\mu).$$

Then

$$(3) \quad \mu(\{x : |\tilde{T}_\varepsilon \sigma(x)| > t\}) \leq c_1 t^{-1} \|\sigma\| \quad \text{for } t > 0$$

and for all complex Radon measures σ on $\mathbf{R}^n$ where c_1 depends only on m , C and c .

Proof. First we observe that (2) yields (3) with $c_1 = c$ for complex Radon measures σ with $\text{spt } \sigma \subset S$. Indeed, for this it suffices to find functions $f_i \in L^1(\mu)$, $i = 1, 2, \dots$, such that $\int |f_i| d\mu \leq \|\sigma\|$ and $\tilde{T}_\varepsilon f_i(x) \rightarrow \tilde{T}_\varepsilon \sigma(x)$ for $x \in \mathbf{R}^n$. It is readily seen that this is achieved for example with

$$f_i = \sum_{Q \in \mathcal{D}_i} \frac{\sigma(Q)}{\mu(Q)} \chi_Q$$

where $\mathcal{D}_i$ is the set of dyadic cubes of side-length 2^{-i} . Now we apply Theorem 20.19 with $L = \tilde{T}_\varepsilon$, $X = Y = S$ and $\nu = \mu$. Then $L^t = -\tilde{T}_\varepsilon$ and we see that for any compact set $F \subset S$ there is a Borel function $h: S \rightarrow [0, 1]$ with $h = 0$ on $S \setminus F$, $\|\tilde{T}_\varepsilon h\|_S \leq 1$ and $7c \int h d\mu \geq \mu(F)$. By Lemma 20.22, $\|\tilde{T}_\varepsilon h\|_{\mathbf{R}^n} \leq c_2$, where c_2 , $1 \leq c_2 < \infty$, depends only on m , C and c . Then $\|\tilde{T}_\varepsilon(h/c_2)\|_{\mathbf{R}^n} \leq 1$ with $\mu(F) \leq 7cc_2 \int (h/c_2) d\mu$ and we can apply Theorem 20.19 to the other direction with $X = S$, $Y = \mathbf{R}^n$ and $\nu = \mu$ to obtain (3) for all complex Radon measures σ on $\mathbf{R}^n$. $\square$

Kolmogorov's inequality

To extend the weak type (1,1) inequality to the maximal operator T^* we need Cotlar's inequality, and for this the following Kolmogorov inequality.

20.24. Lemma. Let $0 < c < \infty$, let ν be a Radon measure on $\mathbf{R}^n$ and $f: \mathbf{R}^n \rightarrow \mathbf{C}$ a Borel function with

$$\nu(\{x : |f(x)| > t\}) \leq ct^{-1} \quad \text{for } t > 0.$$

Then for $0 < s < 1$ and for every ν -measurable set $A \subset X$ with $\nu(A) < \infty$,

$$\int_A |f|^s d\nu \leq c^s (1-s)^{-1} \nu(A)^{1-s}.$$

Proof. Using Theorem 1.15 and denoting $\alpha = c/\nu(A)$ we estimate

$$\begin{aligned} \int_A |f|^s d\nu &= \int_0^\infty \nu(\{x \in A : |f(x)|^s > u\}) du \\ &= s \int_0^\infty t^{s-1} \nu(\{x \in A : |f(x)| > t\}) dt \\ &\leq s \int_0^\alpha t^{s-1} \nu(A) dt + s \int_\alpha^\infty t^{s-1} ct^{-1} dt \\ &= \alpha^s \nu(A) + s(1-s)^{-1} \alpha^{s-1} c = c^s (1-s)^{-1} \nu(A)^{1-s}. \quad \square \end{aligned}$$

Cotlar's inequality

Next we prove a form of Cotlar's inequality involving the truncated maximal functions

$$T_\varepsilon^* \sigma(x) = \sup_{\delta \geq \varepsilon} |T_\delta \sigma(x)|$$

for complex Radon measures σ .

20.25. Lemma. Let $\varepsilon > 0$. Suppose that $c < \infty$ and

$$(1) \quad \mu(\{x : |T_\varepsilon \sigma(x)| > t\}) \leq ct^{-1} \|\sigma\| \quad \text{for } t > 0$$

and for all complex Radon measures σ on $\mathbf{R}^n$. Then for $0 < s \leq 1$ there is $c_s < \infty$ such that for $x \in \mathbf{R}^n$,

$$(2) \quad |T_\varepsilon^* \sigma(x)| \leq c_s (M_\mu(|T_\varepsilon \sigma|^s)(x)^{1/s} + M_\mu \sigma(x)).$$

Proof. Let $\delta \geq \varepsilon$. Write $B = B(x, \delta/2)$ and $\sigma = \sigma_1 + \sigma_2$ with $\sigma_1 = \sigma \llcorner B(x, \delta)$. Then

$$T_\delta \sigma(x) = T_\varepsilon \sigma_2(x).$$

Estimating as in the proof of Lemma 20.22 we find that for some constant c_1 , $c \leq c_1 < \infty$,

$$|T_\varepsilon \sigma_2(x) - T_\varepsilon \sigma_2(y)| \leq c_1 M_\mu \sigma(x)$$

for $y \in B$. Thus for $y \in B$,

$$(3) \quad \begin{aligned} |T_\delta \sigma(x)| &\leq |T_\varepsilon \sigma_2(y)| + |T_\varepsilon \sigma_2(x) - T_\varepsilon \sigma_2(y)| \\ &\leq |T_\varepsilon \sigma(y)| + |T_\varepsilon \sigma_1(y)| + c_1 M_\mu \sigma(x). \end{aligned}$$

Assume first $s = 1$. If $T_\delta \sigma(x) \neq 0$, let $0 < t < |T_\delta \sigma(x)|$. Then by (3) for $y \in B$, either $|T_\varepsilon \sigma(y)| > t/3$ or $|T_\varepsilon \sigma_1(y)| > t/3$ or $c_1 M_\mu \sigma(x) > t/3$. This means that either

$$(4) \quad t < 3c_1 M_\mu \sigma(x)$$

or

$$(5) \quad B = \{y \in B : |T_\varepsilon \sigma(y)| > t/3\} \cup \{y \in B : |T_\varepsilon \sigma_1(y)| > t/3\}.$$

But

$$\mu(\{y \in B : |T_\varepsilon \sigma(y)| > t/3\}) \leq 3t^{-1} \int_B |T_\varepsilon \sigma| d\mu \leq 3\mu(B)t^{-1} M_\mu(T_\varepsilon \sigma)(x)$$

and, by the weak type inequality (1) and 20.12 (2),

$$\begin{aligned} \mu(\{y \in B : |T_\varepsilon \sigma_1(y)| > t/3\}) &\leq 3ct^{-1} \|\sigma_1\| \\ &= 3ct^{-1} \mu(B(x, \delta)) \frac{|\sigma|(B(x, \delta))}{\mu(B(x, \delta))} \leq 3ct^{-1} 2^m C^2 \mu(B) M_\mu \sigma(x). \end{aligned}$$

In both cases (4) and (5) with $c_2 = 3c_1 2^m C^2$,

$$t < 3M_\mu(T_\varepsilon \sigma)(x) + c_2 M_\mu \sigma(x).$$

Since this holds for any $0 < t < |T_\delta \sigma(x)|$ and any $\delta \geq \varepsilon$, (2) follows in the case $s = 1$.

Assume now $0 < s < 1$. Then (3) gives for $y \in B$,

$$|T_\delta \sigma(x)|^s \leq |T_\varepsilon \sigma(y)|^s + |T_\varepsilon \sigma_1(y)|^s + c_1^s M_\mu \sigma(x)^s.$$

Integrating with respect to y and μ over B , dividing by $\mu(B)$ and raising to the power $1/s$, we obtain

$$|T_\delta \sigma(x)| \leq c'_s \left(M_\mu(|T_\varepsilon \sigma|^s)(x)^{1/s} + \left(\mu(B)^{-1} \int_B |T_\varepsilon \sigma_1|^s d\mu \right)^{1/s} + M_\mu \sigma(x) \right).$$

The middle term can be estimated by Kolmogorov's inequality 20.24,

$$\mu(B)^{-1} \int_B |T_\varepsilon \sigma_1|^s d\mu \leq c^s (1-s)^{-1} \mu(B)^{-s} \|\sigma_1\|^s \leq c''_s M_\mu \sigma(x)^s,$$

and (2) follows. $\square$

A weak (1, 1) inequality for complex measures

We now put together the above ingredients to obtain a weak type (1,1) inequality for the maximal operator T^* for general complex Radon measures on $\mathbf{R}^n$.

20.26. Theorem. *Suppose that the operators T_ε , $0 < \varepsilon < \infty$, are uniformly bounded on $L^2(\mu)$ (as in Theorem 20.13). Then there exists $c_1 < \infty$ such that*

$$\mu(\{x : T^*\sigma(x) > t\}) \leq c_1 t^{-1} \|\sigma\| \quad \text{for } t > 0$$

and for any complex Radon measure σ on $\mathbf{R}^n$.

Proof. Since $T^*\sigma(x) = \lim_{\varepsilon \downarrow 0} T_\varepsilon^*\sigma(x)$, it suffices to verify this for T_ε^* with c_1 independent of ε . For this we apply Lemma 20.25. The assumption (1) there holds because of Theorem 20.13, Lemmas 20.21, 20.23 and Theorem 2.19. Choose $s = 1/2$ in Lemma 20.25. Since M_μ is of weak type (1,1) by Theorem 2.19 and $M_\mu \leq \overline{M}_\mu$ (recall 2.18), we are left to show that

$$(1) \quad \mu(A) \leq c_2 t^{-1} \|\sigma\|$$

where

$$A = \{x : \overline{M}_\mu(|T_\varepsilon \sigma|^{1/2})(x)^2 > t\}.$$

This follows from the weak type (1,1) inequality for $\overline{M}_\mu$ in Theorem 2.19 and Kolmogorov's inequality 20.24,

$$\mu(A) \leq c_3 t^{-1/2} \int_A |T_\varepsilon \sigma|^{1/2} d\mu \leq c_4 t^{-1/2} \mu(A)^{1/2} \|\sigma\|^{1/2},$$

which gives (1). □

Rectifiability implies existence of principal values

We can now extend the existence of principal values in Theorem 20.15 to rectifiable sets and general complex Radon measures on $\mathbf{R}^n$.

20.27. Theorem. Let E be an $\mathcal{H}^m$ measurable m -rectifiable subset of $\mathbf{R}^n$ with $\mathcal{H}^m(E) < \infty$. If σ is a complex Radon measure on $\mathbf{R}^n$, then the principal values

$$\lim_{\varepsilon \downarrow 0} \int k_\varepsilon(x - y) d\sigma y$$

exist for $\mathcal{H}^m$ almost all $x \in E$.

Proof. We may assume that E is a compact Lipschitz graph as in Theorem 20.15 since $\mathcal{H}^m$ almost all of E can be covered with countably many such Lipschitz graphs, recall Exercise 15.8. By the Lebesgue decomposition theorem and Radon–Nikodym theorem for complex measures (i.e. the complex analogue of Theorem 2.17), see Rudin [1, 6.9 and 6.10], there exist $f \in L^1(\mathcal{H}^m \llcorner E)$ and a complex Radon measure σ_s such that $|\sigma_s|$ and $\mathcal{H}^m \llcorner E$ are mutually singular and

$$\sigma(B) = \int_{E \cap B} f d\mathcal{H}^m + \sigma_s(B)$$

for Borel sets $B \subset \mathbf{R}^n$. Since the principal values exist $\mathcal{H}^m$ almost everywhere on E for the first summand by Theorem 20.15, we may assume that $\sigma = \sigma_s$.

It suffices to verify the following statement.

For every $\alpha > 0$ there is $\beta > 0$ such that there exists $A_\alpha \subset E$ for which $\mathcal{H}^m(A_\alpha) < \alpha$ and

$$|T_\delta \sigma(x) - T_\varepsilon \sigma(x)| \leq \alpha \quad \text{for } x \in E \setminus A_\alpha, \delta, \varepsilon \in (0, \beta).$$

Then setting $B = \bigcap_{k=1}^\infty \bigcup_{i=k}^\infty A_{2^{-i}}$ we find that $\mathcal{H}^m(B) = 0$ and $\lim_{\varepsilon \downarrow 0} T_\varepsilon \sigma(x)$ exists for $x \in E \setminus B$.

Since E is compact and σ and $\mathcal{H}^m \llcorner E$ are mutually singular, there are for any $\gamma > 0$ an open neighbourhood U of E and a compact subset F of E such that

$$|\sigma|((U \setminus E) \cup (E \setminus F)) < \gamma \quad \text{and} \quad \mathcal{H}^m(F) = 0.$$

As E and F are compact we can choose $\beta > 0$ such that $d(E, \mathbf{R}^n \setminus U) > \beta$ and $\mathcal{H}^m(F_\beta) < \gamma$ where

$$F_\beta = \{x \in E : d(x, F) \leq \beta\}.$$

Let $\tau = \sigma \llcorner ((U \setminus E) \cup (E \setminus F))$. Then $\|\tau\| < \gamma$ and

$$T_\delta \sigma(x) - T_\varepsilon \sigma(x) = T_\delta \tau(x) - T_\varepsilon \tau(x) \quad \text{for } x \in E \setminus F_\beta, \delta, \varepsilon \in (0, \beta).$$

Let

$$A = F_\beta \cup \{x \in E : T^* \tau(x) > \alpha/2\}.$$

Then for $\delta, \varepsilon \in (0, \beta)$ and $x \in E \setminus A$,

$$|T_\delta \sigma(x) - T_\varepsilon \sigma(x)| \leq 2T^* \tau(x) \leq \alpha.$$

By Theorem 20.26,

$$\mathcal{H}^m(A) < \gamma + c\alpha^{-1} \|\tau\| < (1 + c\alpha^{-1})\gamma.$$

We obtain the desired inequality by choosing γ sufficiently small. $\square$

Combining Theorems 20.10 and 20.27 together with Theorem 16.2 (1) we get the following characterization of rectifiability. Recall the definition of the vector-valued kernel K_m and the principal value $K_m \mu(x)$ from (20.3) and (20.4).

20.28. Theorem. (1) Let μ be a finite Radon measure on $\mathbf{R}^n$ with

$$0 < \Theta_*^m(\mu, x) \leq \Theta^m(\mu, x) < \infty \quad \text{for } \mu \text{ almost all } x \in \mathbf{R}^n.$$

Then μ is m -rectifiable if and only if $K_m \mu(x)$ exists for μ almost all $x \in \mathbf{R}^n$.

(2) Let E be an $\mathcal{H}^m$ measurable subset of $\mathbf{R}^n$ with $\mathcal{H}^m(E) < \infty$. Then E is m -rectifiable if and only if $\Theta_*^m(E, x) > 0$ and $K_m(\mathcal{H}^m \llcorner E)(x)$ exists for $\mathcal{H}^m$ almost all $x \in E$.

20.29. Remarks. (1) Theorems 20.26 and 20.27 were first proved by Mattila and Melnikov [1] for the Cauchy kernel $1/z$ in $\mathbf{C}$ with a more direct argument. The above proof is from Verdera [2].

(2) Let $E \subset \mathbf{R}^n$, $E \neq \emptyset$, be closed such that for some positive numbers s , c and d ,

$$(1) \quad cr^s \leq \mathcal{H}^s(E \cap B(x, r)) \leq dr^s \quad \text{for } x \in E \text{ and } r > 0.$$

Vihtilä [1] proved that if the operators $T_\varepsilon = T_\varepsilon^i$ corresponding to the Riesz kernels $x \mapsto x_i |x|^{-s-1}$, $i = 1, \dots, n$, are all uniformly bounded on $L^2(\mathcal{H}^s \llcorner E)$ (as in 20.13 (1)), then s must be an integer. However, as remarked before in 19.18 (5) and 20.16 (2), it is not known if E must be s -rectifiable. If one assumes L^2 -boundedness for all kernels k as in 20.12, then uniform s -rectifiability (recall 15.23) of E follows, see David and Semmes [1]. In fact, using Theorem 20.9 one can show that it is enough to use all the kernels k of the form $k(x) = \varphi(x)|x|^{-m-1}$ where φ is a radial C^∞ function on $\mathbf{R}^n$, see David and Semmes [2, Theorem I.2.59] and Mattila and Preiss [1].

Exercises.

1. Show that if $f: \mathbf{R} \rightarrow \mathbf{R}$ is Hölder continuous and Lebesgue integrable, then the limit $Hf(x)$ as in (20.1) exists for all $x \in \mathbf{R}$.
2. Let ν be a Radon measure on $\mathbf{R}$. Show that ν is symmetric if and only if either $\nu = c\mathcal{L}^1$ for some positive number c , or $\nu = c\delta_x$ for some $x \in \mathbf{R}$ and some positive number c , or $\nu = \sum_{m \in \mathbf{Z}} (a\delta_{x+2my} + b\delta_{x+(2m+1)y})$ for some $x, y \in \mathbf{R}$, $y \neq 0$, and some positive numbers a and b .
3. Let ν be a symmetric measure on $\mathbf{R}^n$ with $0 \in \text{spt } \nu \subset \{x : x_n \geq 0\}$. Show that $\text{spt } \nu \subset \{x : x_n = 0\}$.
4. Show that if $\Gamma \subset \mathbf{R}^2$ is a smooth curve and $\mathcal{H}^1 \llcorner \Gamma$ is symmetric, then Γ is a line.
5. Compute the maximal function H^*f ,

$$H^*f(x) = \sup_{\varepsilon > 0} \int_{\{t: |x-t| > \varepsilon\}} (t-x)^{-1} f(t) dt,$$

at all points $x \in \mathbf{R}$ when $f = \chi_{[0,1]}$.

6. Let $S = C(1/4) \times C(1/4)$, recall 4.10, and $\mu = \mathcal{H}^1 \llcorner S$ for the setting of 20.12 with $k(z) = 1/z$ for $z \in \mathbf{C} \setminus \{0\}$. Show that $T^*\mu(x) = \infty$ for μ almost all $x \in S$. In particular, the operators T_ε are not uniformly bounded on $L^2(\mu)$.